

A coupled bubble plume–reservoir model for hypolimnetic oxygenation

V. L. Singleton,¹ F. J. Rueda,² and J. C. Little¹

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[1] A model for a linear bubble plume used for hypolimnetic oxygenation was coupled with a three-dimensional hydrodynamic model to simulate the complex interaction between bubble plumes and the large-scale processes of transport and mixing. The coupled model accurately simulated the evolution of dissolved oxygen (DO) and temperature fields that occurred during two full-scale diffuser tests in a water supply reservoir. The prediction of asymmetric circulation cells laterally and longitudinally on both sides of the linear diffuser was due to the uneven reservoir bathymetry. Simulation of diffuser operation resulted in baroclinic pressure gradients, which caused vertical oscillations above the hypolimnion and contributed to distribution of plume detrainment upstream and downstream of the diffuser. On the basis of a first-order variance analysis, the largest source of uncertainty for both predicted DO and temperature was the model bathymetry, which accounted for about 90% of the overall uncertainty. Because the oxygen addition rate was 4 times the sediment oxygen uptake (SOU) rate, DO predictions were not sensitive to SOU. In addition to bathymetry, the momentum assigned to plume entrainment and detrainment is a significant source of uncertainty in the coupled model structure and appreciably affects the predicted intensity of mixing and lake circulation. For baseline runs, the entrainment and detrainment velocities were assumed to be half of the velocities through the flux face of the grid cells. Additional research on appropriate values of the plume detrainment momentum for the coupled model is required.

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1. Introduction

[2] Hypolimnetic hypoxia, the depletion of dissolved oxygen (DO) in the bottom water of lakes and reservoirs, can result in severe water quality degradation. Anoxia within lake sediments can increase the rate of soluble phosphorus release [Boström *et al.*, 1982], triggering the growth of nuisance algal blooms [French and Petticrew, 2007]. Oxygen depletion within the hypolimnion may also lead to increases in reduced species such as hydrogen sulfide and the release of soluble iron and manganese from the sediments, which typically require treatment in raw water used for potable purposes [Cooke and Carlson, 1989]. The effects of low DO levels on fish have been studied for decades and include mass mortalities, ecological disruption, physiological stress, and modification of complex behaviors such as schooling, swimming, and reproduction [Pollock and Dubé, 2007]. Methylation of inorganic mercury by microorganisms to a soluble and more bioavailable form is enhanced under anoxic conditions [Mailman *et al.*, 2006]. Last, anoxic water has been shown to greatly increase the

release of major greenhouse gasses such as methane and carbon dioxide from lake sediments [Liikanen *et al.*, 2002].

[3] Bubble plumes are commonly used for hypolimnetic aeration and oxygenation, adding oxygen to the deeper water while preserving stratification [Singleton and Little, 2006]. The plume is in intimate contact with the ambient water and is strongly influenced by the local density gradient. While bubble plumes are successful at adding oxygen, the added energy may induce large-scale hypolimnetic mixing. Plume-induced mixing alters the density structure of the reservoir, while plume performance depends strongly on the density gradient, establishing a feedback loop that continually changes plume behavior [McGinnis *et al.*, 2004]. Mixing may partially erode the thermocline and subsequently lead to warming of the hypolimnion and even premature destratification of the reservoir. Higher hypolimnetic temperatures and plume-induced mixing may also be responsible for increased sediment oxygen uptake (SOU). A number of studies have reported that small increases in water velocity above lake sediments can significantly increase SOU [Arega and Lee, 2005; Bryant *et al.*, 2010; Hondzo, 1998; Josiam and Stefan, 1999; Lorke *et al.*, 2003]. Because the sediment is the largest sink of oxygen in most reservoirs, the effect of plume-induced mixing must be included to avoid serious undersizing of oxygenation systems. The specific plume-induced mixing mechanisms should be identified and incorporated in a coupled bubble plume–reservoir model for successful design and operation.

¹Department of Civil and Environmental Engineering, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, USA.

²Instituto del Agua and Departamento Ingeniería Civil, Universidad de Granada, Granada, Spain.

Table 1. Key Variables of the Linear Bubble Plume Model^a

Variable	Formula	Units
Entrainment	$E = 2(L_{plu} + W_{plu})\alpha v$	$\text{m}^2 \text{s}^{-1}$
Plume water volume flux	$Q_{plu} = L_{plu}W_{plu}v$	$\text{m}^3 \text{s}^{-1}$
Momentum flux	$M = L_{plu}W_{plu}v^2$	$\text{m}^4 \text{s}^{-2}$
Temperature flux	$F_T = Q_{plu}T_p$	$^{\circ}\text{C m}^{-3} \text{s}^{-1}$
Dissolved solids flux	$F_s = Q_{plu}S\rho_w$	kg s^{-1}
Dissolved O ₂ and N ₂ fluxes	$F_{D_i} = Q_{plu}C_i$	mol s^{-1}
Gaseous O ₂ and N ₂ fluxes	$F_{G_i} = \lambda W_{plu}(L_{plu} - W_{plu} + \lambda W_{plu})(v + v_b)y_i$	mol s^{-1}

^aSee Singleton *et al.* [2007].

[4] A number of models for predicting plume dynamics and/or oxygen transfer from bubble plumes have been developed [Singleton and Little, 2006], but less research has focused on modeling the interaction between bubble plumes and ambient water bodies. Most of the existing coupled bubble plume-reservoir models involve whole-lake artificial circulation or destratification and do not account for mass transfer from the bubbles [Hodges and Dallimore, 2006; Imerito, 2007; Johnson *et al.*, 2000; Schladow, 1993; Zic and Stefan, 1994]. Estimating oxygen transfer is critical to predicting the performance of hypolimnetic oxygenation systems. Furthermore, all of the models except one (ELCOM) assume homogeneous conditions in the horizontal direction, ignoring the fact that near-field conditions determine the operational behavior of bubble plumes, as shown by McGinnis *et al.* [2004]. Even though ELCOM is three dimensional, use of the model for destratification has not been validated in the literature.

[5] McGinnis *et al.* [2004] conducted a detailed analysis of the plume-lake interaction in Lake Hallwil, Switzerland. Entrainment occurred from the near field, which was where detained water accumulated over tens of meters from the plume, as opposed to the far field as previously thought. Because entrainment was mostly from the near field, plume temperature and constituent concentrations depended strongly on that region, which was created by the plume itself and continually evolving. Currently, near-field boundary conditions cannot be predicted for bubble plume design and must either be measured or assumed. Bravo *et al.* [2007] used a comprehensive three-dimensional model, FLUENT, and constructed a two-fluid dispersed model to simulate the dynamics of bubble plumes. The extension of this model to large-scale and long-term simulations, though, is not straightforward and is computationally expensive.

[6] To address these shortcomings, this work presents the coupling of a linear bubble plume model that includes gas transfer with a three-dimensional hydrodynamic and transport model for long-term basin-scale simulations in lakes and reservoirs. The coupled models are used to predict plume dynamics, induced mixing, and oxygen addition from a linear diffuser used for hypolimnetic oxygenation. In this paper, the components of the coupled model and the coupling procedure are described, field data from a full-scale system are compared to model predictions, a sensitivity analysis of the coupled model is presented, and sources of model uncertainty are discussed.

2. Methods

2.1. Approach

[7] A model capable of simulating the complex interaction between bubble plumes and basin-scale hydrodynamic

processes has been developed by combining a three-dimensional (3-D) hydrodynamic model of lake transport and mixing [Smith, 2006] and a steady one-dimensional (1-D) linear bubble plume model presented by Singleton *et al.* [2007], which is based on the circular bubble plume model originally proposed by Wüest *et al.* [1992]. The particular procedures adopted to couple the 1-D steady state bubble plume model and the 3-D hydrodynamic model are outlined and justified in this section.

2.2. Linear Bubble Plume Model

[8] The linear bubble plume model was recently validated by Singleton *et al.* [2007] using full-scale diffuser data collected from Spring Hollow Reservoir (SHR), Virginia, United States. The model uses the discrete-bubble approach, which has been successfully applied to other hypolimnetic oxygenation devices and was reviewed in detail by Singleton and Little [2006]. The linear bubble plume model is derived from the circular model of Wüest *et al.* [1992] and is composed of horizontally integrated equations based on the conservation of mass, momentum, and heat. Eight flux equations (Tables 1 and 2) are solved simultaneously to predict water flow rate, temperature, oxygen and nitrogen transfer and concentration, salinity, and rise height, given diffuser geometry and depth, applied gas flow rate, and initial bubble size [Singleton *et al.*, 2007]. The model accounts for vertical density gradients due to temperature and salinity. Entrainment is assumed to be proportional to the local (with respect to depth) plume water velocity. Bubble size varies as the bubbles rise due to expansion and dissolution, and bubble slip velocity and gas transfer coefficients are functions of bubble radius [Wüest *et al.*, 1992]. Henry's constants for oxygen and nitrogen are functions of temperature [Wüest *et al.*, 1992]. The bubble plume model equations were originally developed by Wüest *et al.* [1992] for circular geometry, but were modified by McGinnis *et al.* [2001] for the linear geometry of the system installed in SHR. Singleton *et al.* [2007] improved the linear bubble plume model by use of a correlation to calculate initial bubble size [McGinnis and Little, 2002], correction of the entrainment coefficient (α) and spreading coefficient (λ) for top hat profiles [Fanneløp and Sjøen, 1980], and use of a Froude number (Fr) to calculate initial water velocity

Table 2. Nonlinear Differential Flux Equations of the Linear Bubble Plume Model^a

Variable	Formula
Water volume flux	$\frac{dQ_{plu}}{dz} = E$
Momentum flux	$\frac{dM}{dz} = \frac{\rho_a - \rho_w}{\rho_p} g L_{plu} W_{plu} + \frac{\rho_w - \rho_p}{\rho_p} g \lambda W_{plu} \cdot [L_{plu} - W_{plu}(1 - \lambda)]$
Temperature flux	$\frac{dF_T}{dz} = E T_a$
Salinity flux	$\frac{dF_s}{dz} = E \rho_a S_a$
Dissolved gas flux	$\frac{dF_{D_i}}{dz} = E C_a + \frac{4\pi r^2 N}{v + v_b} K_L (H_i P_i - C_i)$
Gas flux	$\frac{dF_{G_i}}{dz} = -\frac{4\pi r^2 N}{v + v_b} K_L (H_i P_i - C_i)$

^aSee Singleton *et al.* [2007].

[Fischer *et al.*, 1979; Wüest *et al.*, 1992]. Except as noted previously, initial conditions for the bubble plume model were determined as detailed by Wüest *et al.* [1992]. The α and λ were set at 0.11 and 0.93, respectively [Fannelop *et al.*, 1991]. The initial Fr for the linear bubble plume was taken as 2.0 [Singleton *et al.*, 2007]. Ambient vertical profiles of temperature and DO were used for plume boundary conditions and salinity was not modeled. Additional details about the bubble plume model are given by Singleton *et al.* [2007].

[9] For this work, the plume model was only applied over the plume rise height up to the first detrainment depth, which is referred to hereafter as the depth of maximum plume rise (DMPR). For hypolimnetic oxygenation diffusers, the depth of maximum plume rise typically corresponds to the top of the hypolimnion. When the plume stops rising, a secondary plume may form above as bubbles that are not completely dissolved continue to rise [Asaeda and Imberger, 1993; McDougall, 1978; Schladow, 1992]. The formation of plumes above the first detrainment depth was not considered in this analysis.

2.3. Hydrodynamic Transport and Mixing Model

[10] The three-dimensional free surface hydrodynamic model used in the coupling exercise has been extensively validated against analytical solutions and field observations collected in a number of lakes [Rueda and Schladow, 2003; Rueda and Cowen, 2005]. The model is based on the continuity equation for incompressible fluids, the Reynolds-averaged form of the Navier-Stokes equations for momentum, and the transport equation for temperature. The governing equations are first posed in layer-averaged form by integrating over the height of a series of horizontal layers separated by level planes. In layer-averaged form, the layer-integrated volumetric transports replace velocities as the dependent variables, so that the depth-integrated continuity equation that is used in the solution for the water surface elevation is linear. This approach helps to ensure that mass and momentum are strictly conserved in the model [Smith, 2006]. The layer-averaged momentum equations are solved using a semi-implicit, three-level, iterative leapfrog-trapezoidal finite difference scheme on a staggered Cartesian grid. Turbulent mixing is represented in the 3-D model using diffusion-like terms. A Laplacian operator with constant mixing coefficients (horizontal eddy viscosity A_h or diffusivity D_h) is used in the model to represent horizontal mixing of momentum and scalars (see equations (2)–(5)). The quasi-equilibrium turbulent energy model of Galperin *et al.* [1988], based on the Mellor and Yamada level 2.5 turbulence closure [Mellor and Yamada, 1974] and modified by Kantha and Clayson [1994], is used to calculate the vertical turbulent mixing coefficients A_v and D_v . The discrete forms of the two governing equations existing in the turbulence submodel, one for turbulent kinetic energy and another for an eddy length scale, are only solved during the first iteration during each time step [Rueda *et al.*, 2005]. It is from the estimates of turbulent kinetic energy and length scales that A_v is calculated. The scalar transport equations are solved using a two-level, operator-splitting and semi-implicit scheme, in which only vertical diffusion is discretized implicitly. The 1-D advection terms in the transport equation are discretized with flux limiter methods [Durran,

1999]. The corrected fluxes are constructed with the monotone upstream differencing scheme, the Lax-Wendroff second-order method, and the Van Leer MC limiter. Additional details of the hydrodynamic model can be found in [Smith, 2006].

2.4. Coupling Procedures

[11] Operation of the linear bubble plume in the reservoir is simulated in the hydrodynamic model as a series of water columns in which water is relocated from the lower cells, where ambient water is entrained into the plume, into the cells at the top of the plume (the DMPR) where oxygenated water is detrained. Relocation of water is simulated by considering sources and sinks in each water column that contains the linear diffuser. The governing equations solved by the coupled bubble plume-hydrodynamic model, incorporating source-sink terms, are given by (adapted from the work of Lynch [1986])

$$\frac{\partial \zeta}{\partial t} + \frac{\partial}{\partial x} \left[\int_{-H}^{\zeta} u dz \right] + \frac{\partial}{\partial y} \left[\int_{-H}^{\zeta} v dz \right] = 0 \quad (1)$$

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - fv = & - \left(g \frac{\partial \zeta}{\partial x} + g \frac{1}{\rho_0} \int_z^{\zeta} \frac{\partial \rho}{\partial x} dz' \right) \\ & + \frac{\partial}{\partial x} \left(A_h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_h \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) + \frac{S}{\rho_0} (u - u_0) \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + fu = & - \left(g \frac{\partial \zeta}{\partial y} + g \frac{1}{\rho_0} \int_z^{\zeta} \frac{\partial \rho}{\partial y} dz' \right) \\ & + \frac{\partial}{\partial x} \left(A_h \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_h \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) + \frac{S}{\rho_0} (v - v_0) \end{aligned} \quad (3)$$

$$\begin{aligned} \frac{\partial \theta}{\partial t} + u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} + w \frac{\partial \theta}{\partial z} = & D_h \frac{\partial^2 \theta}{\partial x^2} + D_h \frac{\partial^2 \theta}{\partial y^2} + \frac{\partial}{\partial z} \left(D_v \frac{\partial \theta}{\partial z} \right) \\ & + \frac{H}{\rho c_p} + \frac{S}{\rho_0} \theta_0 \end{aligned} \quad (4)$$

$$\frac{\partial O}{\partial t} + u \frac{\partial O}{\partial x} + v \frac{\partial O}{\partial y} + w \frac{\partial O}{\partial z} = D_h \frac{\partial^2 O}{\partial x^2} + D_h \frac{\partial^2 O}{\partial y^2} + \frac{\partial}{\partial z} \left(D_v \frac{\partial O}{\partial z} \right) + \frac{S}{\rho_0} O_0 \quad (5)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \frac{S}{\rho_0} \quad (6)$$

[12] Here the free surface elevation is denoted by ζ ; u , v , and w represent the velocity components in the x , y , and z directions, θ is temperature, O is dissolved oxygen concentration, and H is a source of heat associated with heat and energy fluxes due to atmospheric heating or cooling. The density ρ is calculated from temperature using an equation of state. The subscript 0 is intended to define the characteristics of the water being added or removed from the computational domain at a source or sink cell. S denotes the fluid source strength, and has units of $M L^{-3} T^{-1}$. The ratio S/ρ_0 , for a given computational source cell of nominal

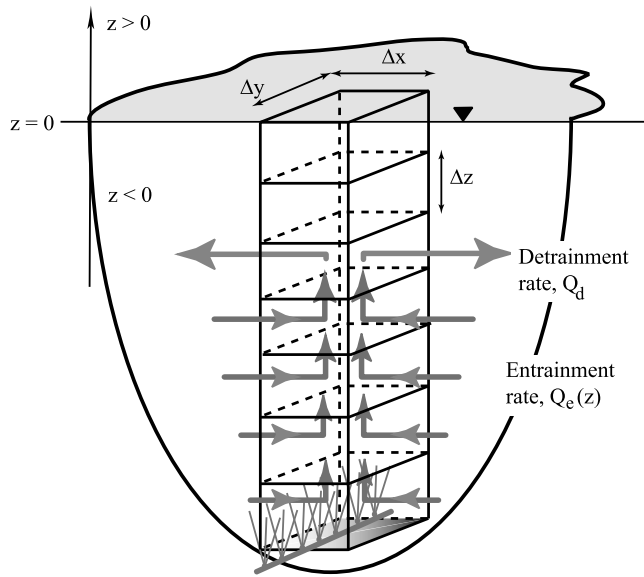


Figure 1. Schematic of the coupling between the 1-D linear bubble plume model and the 3-D hydrodynamic model showing entrainment and detrainment cells.

volume $V = (\Delta x \times \Delta y \times \Delta z)$, represents the volume of water added per unit time, divided by the nominal volume of the cell. Note that no source terms are included in the depth-integrated form of the continuity equation (equation (1)). This approximation is valid in simulating the effects of bubble plumes, since all water entrained into the rising plume reappears in the detrainment cell at the top of the plume. Hence, water is only relocated within the water column, and no sources or sinks exist when considering the entire water column.

[13] The steady state bubble plume model estimates the DMPR, the flow rate of ambient water entrained as the plume rises, and the flow rate of water that is detrained upon reaching the DMPR for a given ambient condition (in particular, temperature), which is predicted by the dynamic reservoir model. The coupling is performed on a water column by water column basis for each diffuser segment (Figure 1). The hydrodynamic model provides profiles of temperature and DO that are used as boundary conditions by the bubble plume model. The bubble plume model then predicts the detrainment flow rate Q_d and the cell into which water is discharged. This cell is referred to as the detrainment cell (DC) and is unique in each column. For simplicity, plume spreading is not considered and the plume width is assumed to occupy a single column. The planar dimensions of each column in the grid for SHR are $15 \text{ m} \times 15 \text{ m}$, and the predicted plume width in SHR was less than 15 m except for the top 1–2 m of the plume. For the DC, the source strength terms are positive and defined as Q_d/V , where V is the nominal cell volume. All cells below the DC within a column are considered entrainment cells (ECs). The flow rates being entrained into the plume at the ECs are calculated by the plume model, and in general vary with vertical position, $Q_e(z)$. The coupled model assumes that detrainment occurs only at the detrainment cell. The temperature of the water added to the DC is the flow-weighted average of the temperatures in the ECs in the column below. The DO concentration of the water detrained to the DC is the sum of

(1) the flow-weighted average of the DO concentrations of the ECs and (2) the oxygen mass addition rate through the diffusers divided by Q_d . This method ensures that mass and heat are conserved during diffuser operation. Additionally, the oxygen transfer efficiency of the linear diffuser was assumed to be 100 percent (this assumption is valid for the full-scale test cases that were analyzed in this work). We will initially assume that the entrained or detrained water has a velocity equal to $U_f/2$, where U_f is the entrainment/detrainment flow rate divided by the area of the flux faces (indicated by arrows in Figure 1). The values of U_f for each entrainment and detrainment cell are calculated by the plume model and account for the effect of gas flow rate on plume rise velocity and, subsequently, entrainment velocity. We will discuss the effects of other choices for u_0 and v_0 in the sensitivity analysis.

[14] Information is exchanged between the linear bubble plume model and the reservoir model, and the linear bubble plume model is executed at time intervals of Δt_c which is not necessarily equal to the hydrodynamic time step Δt . The latter is typically on the order of $O(10)$ s for computational grids with cells of horizontal dimensions of $O(10)$ m. The former is dependent on the estimated time that ambient conditions in the near field of the plume (i.e., boundary conditions) will change significantly. That time scale T_c can be estimated as the time it takes to replace the volume of water existing in an entrainment cell V at any given time, if the flow rate is equal to the entrainment flow rate Q_e , i.e., $T_c = V/Q_e$. For typical values of $V \approx O(10^2) \text{ m}^3$, which corresponds to cell dimensions of $\Delta x = \Delta y = O(10) \text{ m}$ and $\Delta z = O(1) \text{ m}$; and $Q_e \approx O(10^{-1}-10^0) \text{ m}^3 \text{ s}^{-1}$, $T_c = O(10^2-10^3) \text{ s}$.

[15] The use of a quasi steady state model to represent the plume dynamics in a reservoir hydrodynamic and transport model is justified in that the characteristic time scale of the evolution of plume properties T_{plu} is short compared with the time scale of evolution of the basin-scale fields T_{bas} , i.e., $T_{plu} \ll T_{bas}$. The scale T_{plu} can be estimated as H/v_b where H is the DMPR and v_b is the bubble rise velocity. The scale T_{bas} , in turn, can be estimated as the length of time a perturbation in the temperature field (as induced by the plume) propagates to the ends of the basin. If we assume that the diffuser operates at the center of the basin, then $T_{bas} = 0.5 L_{lake}/v_{igw}$, where v_{igw} is the velocity of propagation of internal gravity waves. For an average bubble rise velocity of $v_b = 0.23 \text{ m s}^{-1}$ [Wüest et al., 1992], $v_{igw} = 0.2 \text{ m s}^{-1}$ [Imboden and Wüest, 1995], $H = O(10) \text{ m}$, and $L_{lake} = O(10^2-10^3) \text{ m}$ (typical of lake environments), $T_{bas} = O(10^3-10^4) \text{ s}$, and $T_{plu} = O(10^2) \text{ s}$, 1–2 orders of magnitude smaller.

2.5. Data Collection

[16] To evaluate the performance of the coupled bubble plume-reservoir model, experimental data were collected in Spring Hollow Reservoir, where a linear diffuser is in operation (Figure 2). Constructed in 1995, SHR is a small monomictic, mesotrophic side-stream reservoir that is generally stratified from May to December. The reservoir is managed by the Western Virginia Water Authority and serves as one of the drinking water sources for Roanoke County. The water body has a maximum depth of 65 m and a maximum surface elevation of 431 m. The approximate surface area and volume at full pool are 0.54 km^2 and $12.4 \times 10^6 \text{ m}^3$, respectively. The reservoir is supplied with water pumped

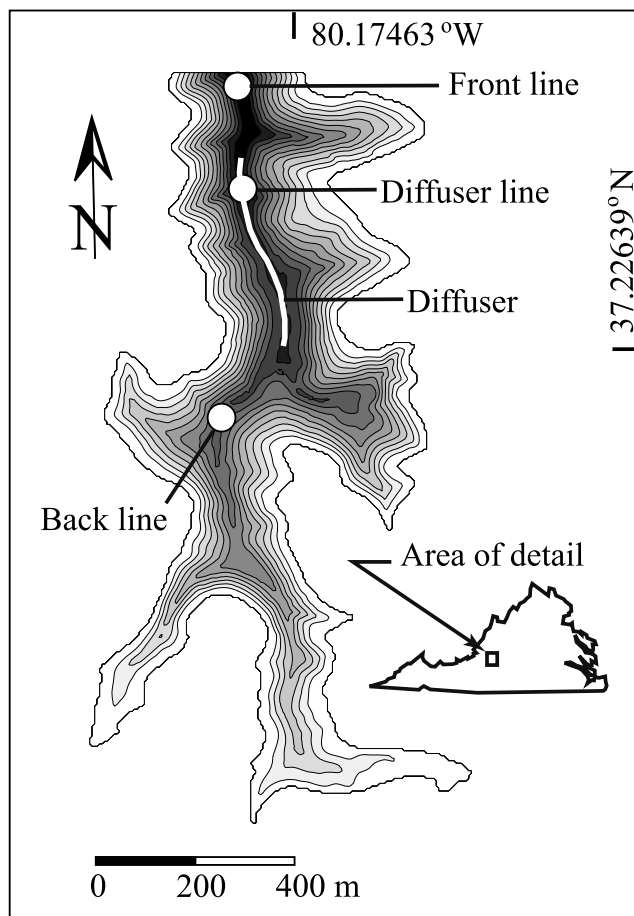


Figure 2. Bathymetric map of Spring Hollow Reservoir, Virginia, United States, with isobaths every 5 m. The white solid line represents the location of the linear bubble plume diffuser in 1998. The sites where the CTD profiles were collected in 2003 are represented as white circles.

from the Roanoke River. The pumped inflow enters the reservoir through a vertical standpipe that discharges at 396 m elevation about 700 m upstream of the dam. To prevent anoxia in the hypolimnion and the associated deterioration of water quality, a linear diffuser equipped with fine-bubble porous hoses was installed in 1997 (Figure 2). The diffuser can be supplied with compressed air or pure oxygen at various gas flow rates and is located in the deepest portion of the reservoir (368–372 m elevation). The diffuser is designed to release a uniform gas flow rate along its length.

[17] A series of diffuser tests was performed in 1998 and 2003 [McGinnis *et al.*, 2001; Singleton *et al.*, 2007]. Compressed air (21% O_2) at a high gas flow rate ($45 \text{ Nm}^3 \text{ h}^{-1}$ average) was supplied to the diffuser from 28 September to 13 October in 1998. Pure oxygen (97% O_2) was, in turn, supplied at a low flow rate ($23 \text{ Nm}^3 \text{ h}^{-1}$ average) from 14 to 26 August 2003. Table 3 provides the diffuser operating parameters during the 1998 and 2003 tests. The starting water surface elevations during the 1998 and 2003 diffuser tests were 420.1 and 430.1 m, respectively.

[18] Data collected during the two diffuser tests were used to evaluate the coupled model (Table 3). During the 1998 test, no water was pumped from the river into the reservoir during diffuser operation, thereby simplifying reservoir

hydrodynamics. The primary hydraulic forcing in the hypolimnion during testing was the linear bubble plume. Because of equipment limitations, temperature and DO profiles were obtained from only a single location adjacent to the dam, approximately 155 m downstream of the end of the diffuser (Figure 2). Profiles were measured immediately before the diffuser was started on 28 September and on four additional days during diffuser operation.

[19] To establish appropriate boundary conditions for the bubble plume model, characterization of the plume near-field environment is necessary [McGinnis *et al.*, 2004]. Therefore, the data collected in 2003 included numerous high spatial resolution CTD (Sea-Bird Model SBE 19plus; 4 Hz sampling rate) transects measured almost daily before, during, and after diffuser operation. The CTD profiler was also equipped with a DO probe (1.4 s response time measured at 20°C). Profiles were obtained laterally across the midpoint of the diffuser (at the diffuser line location) and at several locations longitudinally along the thalweg of the reservoir (Figure 2, at the front, diffuser, and back line locations).

2.6. Model Setup

[20] The coupled bubble plume-reservoir model was applied to simulate the mixing and transport processes occurring in Spring Hollow Reservoir during the two oxygenation tests. The lake geometry was discretized using $15 \times 15 \times 0.5 \text{ m}$ grid cells, and the time step was set to 10 s for stability purposes. With this grid resolution, the advective terms in our model should be able to resolve transport features of $O(10^2) \text{ m}$, the scale of the eddies which would contribute most to the horizontal mixing of the reservoir [Knauer *et al.*, 2000]. The horizontal eddy viscosity A_h and diffusivity D_h in the model were set to $10^{-2} \text{ m}^2 \text{ s}^{-1}$. Use of these values for horizontal diffusivities are supported by results from tracer release experiments by Peeters *et al.* [1996] in the metalimnion of small stratified lakes, the regression curve proposed by Lawrence *et al.* [1995] on the basis of their own observations, and those of Okubo [1971] and Murthy [1976]. The 95% confidence interval in the regression of Lawrence *et al.* [1995] includes values for D_h ranging from $4 \times 10^{-3} \text{ m}^2 \text{ s}^{-1}$ to $2 \times 10^{-2} \text{ m}^2 \text{ s}^{-1}$. Model runs were conducted by varying A_h and D_h within this confidence interval, and the results were not sensitive to differences in A_h and D_h of this magnitude. With horizontal diffusivities of the order $O(10^{-2}) \text{ m}^2 \text{ s}^{-1}$, it is expected that diffusion terms, representing nonresolved subgrid-scale mixing processes, will have negligible effects on the horizontal spreading of patches during the 10–20 day period that was simulated for each diffuser test. For time scales of 10 days, the horizontal spreading of any patches induced by the oxygenation system will be mainly driven by advection processes. With $D_h = O(10^{-2}) \text{ m}^2 \text{ s}^{-1}$ for a 10 day period, only patches with a horizontal scale of $(D_h t)^{1/2} = O(10^2) \text{ m}$ would be homogenized.

[21] The diffuser was modeled as discrete segments at the bottom of consecutive columns along the thalweg in the reservoir model grid. This allowed the elevation of each segment to vary along the reservoir bottom. The number of plume entrainment cells per column varied depending on diffuser depth and predicted plume rise height within the column. Detrainment was assumed to occur from a single cell at the DMPR.

Table 3. Conditions for Coupled Hydrodynamic–Bubble Plume Model Validation and Sensitivity Analysis for Spring Hollow Reservoir, Virginia, United States^a

Parameter	September 1998	August 2003	September 1998 Sensitivity Range
Oxygen in gas supply (%)	21	97	NA
Entrainment coefficient	0.11	0.11	NA
Spreading coefficient	0.93	0.93	NA
Initial Froude number	2.0	2.0	NA
Operational diffuser length ^b (m)	360	145	330–390
Initial plume width (m)	0.16	0.16	NA
Initial plume area (m ²)	58	23	53–62
Gas flow rate ^c (Nm ³ h ⁻¹)	43	23	39–47
Oxygen loading rate ^d (g s ⁻¹)	3.53	8.65	3.18–3.88
Initial gas flux ^e (m h ⁻¹)	0.74	1.0	0.67–0.82
Initial bubble radius (m)	5.7×10^{-4}	5.6×10^{-4}	5.13×10^{-4} – 6.27×10^{-4}
Diffuser depth ^f (m)	50	60	NA
SOU (g m ⁻² d ⁻¹)	0.6	1.5	0.54–0.66
Velocity of detrainment	$U_f/2$	$U_f/2$	U_{amb} and U_f
Reservoir maximum depth (m)	54	65	NA
Reservoir surface area (10 ⁶ m ²)	0.38	0.53	NA
Reservoir total volume (10 ⁶ m ³)	8.2	12	NA

^aReservoir conditions during testing are also included. NA means not applicable.

^bOperational diffuser length varied as a function of gas flow rate because linear diffuser in SHR is designed to have a uniform gas flow rate per unit length.

^c1 Nm³ denotes 1 m³ of gas at 1 bar and 0°C.

^dAssumes 100 percent oxygen transfer efficiency.

^eSensitivity analysis range refers to varying gas flow rate while maintaining constant initial plume area.

^fDepth at location of lateral CTD transect for 2003. For 1998, depth corresponds to about 230 m upstream of the dam.

[22] The initial boundary conditions for the bubble plume model were equal to the initial conditions selected for the hydrodynamic model, as detailed in the next paragraph. Subsequent boundary conditions for the plume model were as predicted by the reservoir model for the columns laterally adjacent to the columns which contained the diffuser. As the runs progressed, plume boundary conditions were updated every 30 min.

[23] Initial conditions for the hydrodynamic model included temperature and DO profiles. The entire reservoir was initialized to a single grid-wide vertical profile for each constituent. For 1998 and 2003, the initial reservoir conditions were equal to the profiles measured immediately before diffuser start-up. Because profiles were collected at numerous locations in 2003, the initial conditions for that diffuser test were the profiles collected at the midpoint of the diffuser (diffuser line location, Figure 2). (Note that the diffuser was repositioned closer to the dam in 2002.) In general, if vertical profiles from multiple locations are available to initialize the reservoir model, it is recommended that profiles closest to the plume near field (± 1 – 2 m laterally from diffuser centerline) be used [McGinnis *et al.*, 2004]. If multiple near-field profiles have been collected prior to diffuser start-up, then profiles closest to the longitudinal midpoint should be used for initial conditions.

[24] The hydrodynamic model was forced using (1) observed flow rate and temperature data for the pumped raw water inflow entering the reservoir (inflow boundary conditions), (2) measured outflow rates to the water treatment plant through a withdrawal gate located at 396 m elevation (outflow boundary conditions), (3) surface heat and momentum fluxes (surface boundary conditions) estimated from meteorological records collected either at the local airport (for 1998) or by a weather station installed on the lake (for 2003), and (4) operational data from the linear bubble plume diffusers (Table 3). Other inflows and outflows, such as

precipitation, evaporation, and seepage, were not included in the model given that their contribution to the water balance during the short time periods of the simulations was minimal compared to pumped inflows or outflows to the treatment plant. Standard bulk aerodynamic formulae [see Rueda, 2001, and references therein] were used to calculate surface heat and momentum fluxes from meteorological records. The DO concentration of the pumped inflow was estimated using the measured temperature of the flow and assuming 100 percent saturation. The outflow to the treatment plant was the largest measured flow rate from the reservoir. The oxygen simulations included transport, sources and sinks above the diffuser line representing the entrainment of ambient water at the base of the plumes and the detrainment of oxygenated water at the top of the plume, and a sink term in the bottom cells representing SOU. For 1998, the SOU was determined by applying a mass balance on DO in the hypolimnion during the diffuser operating period and assuming that the sediments were the only sink for hypolimnetic DO depletion. For 2003, the SOU was estimated using the correlation of Gantzer *et al.* [2009], which was developed using DO data collected from SHR during 2003–2008 and is a function of gas flow rate. The correlation actually calculates the hypolimnetic oxygen demand (HOD), or oxygen demand in both the water column and sediments, induced by diffuser operation. However, because data on SOU are not available for SHR, we assume that HOD is equivalent to SOU in this work. The SOU rates used in the simulations are summarized in Table 3.

2.7. Sensitivity Analysis

[25] A sensitivity analysis was performed with the coupled reservoir–bubble plume model to determine the effects of select model parameters and model configuration modes (Table 3) on the predicted DO and temperature fields in the

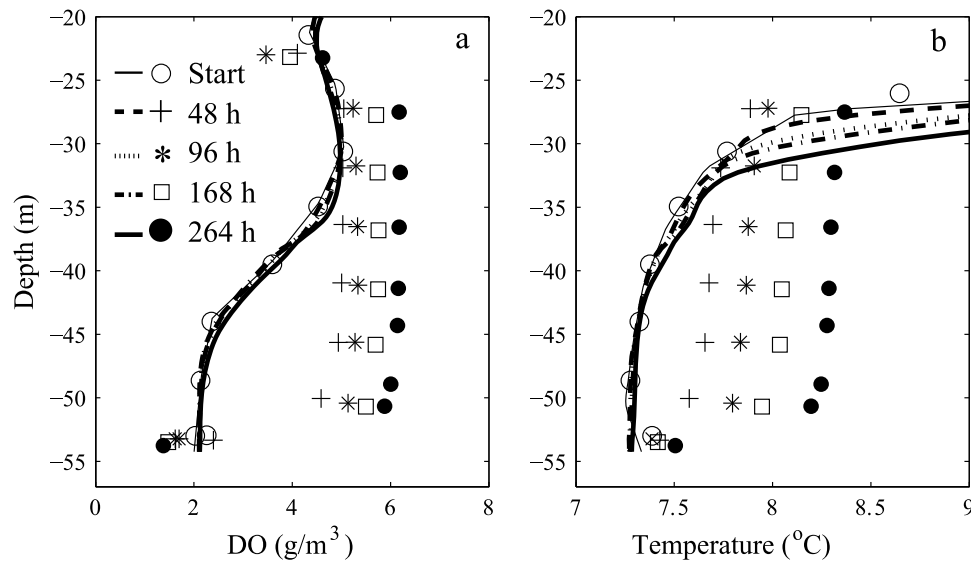


Figure 3. Hypolimnetic profiles predicted by 3-D hydrodynamic model alone, represented as solid and dashed lines. Observed (a) DO and (b) temperature are represented as symbols. Data were collected 8 m upstream of dam in Spring Hollow Reservoir, Virginia, United States, during diffuser operation in September 1998.

hypolimnion. We adopted the first-order variance analysis (FOVA), as outlined by *Blumberg and Georgas* [2008], and the associated dimensionless sensitivity coefficients to quantify model sensitivity. The dimensionless sensitivity coefficient S_p of any model output variable F (for example, DO concentration) to perturbations in any given parameter or forcing variable p can be determined numerically by the FOVA as

$$S_p(p_0) = \frac{\Delta F / F(p = p_0)}{\Delta p / p_0} \quad (7)$$

where ΔF is the change in the output variable that results from a sufficiently small ($<10\%$) change or perturbation in the parameter Δp , from a reference or baseline value p_0 . The sensitivity coefficient can be interpreted as the percentage change in the output variable F resulting from a 1% uncertainty in the model parameter p . This approach can be repeated one at a time for any given set of parameters, providing a relatively simple and straightforward alternative to other techniques that have been proposed in the literature for assessing the sensitivity of hydrodynamic models [*Piasecki and Katopodes*, 1999; *Salas and Shin*, 1999; *Sanders and Katopodes*, 2000]. To calculate each dimensionless sensitivity coefficient, $\Delta F / F$ was evaluated as the L1 norm [*Trefethen and Bau*, 1997] of the difference between the output variable evaluated at the perturbed $F(p_0 + \Delta p)$ and nonperturbed $F(p_0)$ values of the parameter p . The L1 norm was also used to evaluate the magnitude of $\Delta p / p_0$ in cases where the input variable varied spatially. Once S_p is calculated for a range of parameters, a normalized measure of the output uncertainty $|S|$ can be estimated as

$$|S| = \sqrt{\sum_{i=1}^n S_{p(i)}^2} \quad (8)$$

where n is the number of parameters being considered in the sensitivity analysis, and $S_{p(i)}$ are the dimensionless sensitivity coefficients for the parameter $p(i)$. One can also calculate the

relative contribution R_i of each of the assessed input parameters to the total measure of output uncertainty as

$$R_i = \frac{S_{p(i)}^2}{|S|^2} \quad (9)$$

Equation (9) can then be used to rank the contribution of each input parameter, assuming that each is known within the same uncertainty range, to the total resulting uncertainty in F .

3. Results and Discussion

3.1. Effect of Coupling on Temperature and Dissolved Oxygen

[26] To demonstrate that coupling of the bubble plume model with the hydrodynamic model does indeed improve the prediction of temperature and dissolved oxygen, the coupled model was run with the diffuser turned off (i.e., the hydrodynamic model alone). The predicted hypolimnetic profiles did not vary much from the initial conditions during the course of the 11 day diffuser test (Figure 3). By contrast, coupling of the bubble plume and hydrodynamic models resulted in greatly improved predicted profiles in the hypolimnion (Figure 4). The decreasing temperature gradient at the top of the hypolimnion predicted by the hydrodynamic model alone (Figure 3b) is likely due to wind mixing at the surface, evidenced by cooling and deepening of the epilimnion observed during the diffuser test (not shown).

3.2. Coupled Model Validation

[27] The coupled model was used to predict the evolution of temperature and DO within the hypolimnion for September 1998 (Figure 4). Diffuser operation resulted in more homogeneous vertical temperatures and increased the bulk hypolimnion temperature, due to entrainment of metalimnetic water. The vertical distribution of oxygen is almost uniform throughout the hypolimnion; the large DO concentrations in the core of the hypolimnion due to diffuser operation rapidly

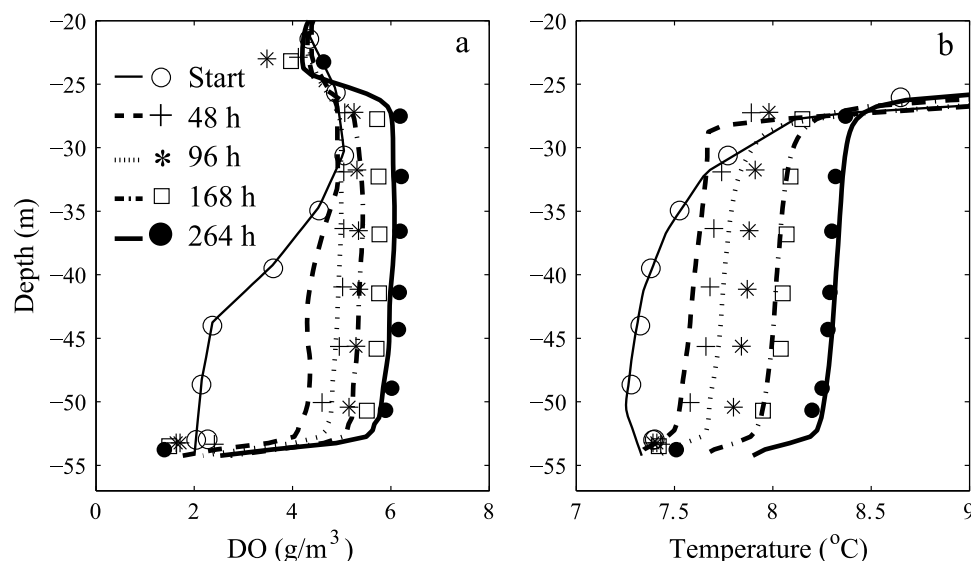


Figure 4. Hypolimnetic profiles predicted by coupled bubble plume–3-D hydrodynamic model, represented as solid and dashed lines. Observed (a) DO and (b) temperature are represented as symbols. Data were collected 8 m upstream of dam in Spring Hollow Reservoir, Virginia, United States, during diffuser operation in September 1998.

decrease toward the metalimnion and toward the sediments. The former boundary layer is the result of limited vertical mass exchange at the thermocline. The bottom boundary layer is caused by the sediments removing oxygen from the near-sediment water at a faster rate than it can be replenished by vertical mixing or lateral circulation.

[28] Overall, the coupled model predicts the bulk DO and temperature profiles quite well for September 1998 (Figure 4). Bulk hypolimnion temperatures were also well simulated (Figure 4b). Underprediction of temperature at 48 h and 96 h after start-up could have been caused by underestimated horizontal mixing. Predicted bottom temperatures and DO concentrations are greater than observed, which suggests that the model slightly overestimates vertical mixing rates. Possible reasons for this include inaccuracies in the bathymetry representation, horizontal dispersion coefficient, or induced plume water flow rate.

[29] The coupled bubble plume–hydrodynamic model was also used to predict the hypolimnetic DO and temperature at the three longitudinal locations along the reservoir thalweg for August 2003 (Figure 5). The coupled model accurately predicted the bulk DO increase at each location over the course of the diffuser test. However, the modest hypolimnetic warming caused by diffuser operation was slightly overestimated for the location downstream of the diffuser (Figure 5f). This suggests that diffuser-induced vertical mixing, and subsequent erosion of the thermocline, is overpredicted longitudinally away from the diffuser, similar to the results for the 1998 diffuser test. Additional evidence of overestimated mixing are the overly strong temperature and DO gradients predicted at the top of the hypolimnion for the downstream location. Even though vertical mixing is overestimated downstream of the diffuser, the coupled model quite accurately predicted hypolimnetic warming above the midpoint of the diffuser (Figure 5d).

[30] Some of the discrepancies between observed and predicted profiles at the thalweg locations downstream (Figures 5e and 5f) and upstream (Figures 5a and 5b) of the linear diffuser in August 2003 are due to inaccurate initial boundary conditions. Spatially varying initial conditions, though, are difficult to determine from the available field data. Consequently, uniform conditions are commonly presumed in three-dimensional lake modeling [e.g., *Hodges et al.*, 2000; *Rueda et al.*, 2003b]. For the 2003 simulations, the reservoir initial conditions were assumed to be equal to the measured temperature and DO profiles at the midpoint of the diffuser (Figures 5c and 5d, start). Even though the initial DO gradient specified above the sediments is much weaker than observed for the location furthest upstream of the dam (Figure 5a), the final predicted DO gradient in the bottom boundary layer is stronger and resembles the shape of the measured profile. A single value for SOU was determined for each diffuser test, and the SOU for 2003 was calculated using a correlation which is a function of diffuser gas flow rate [*Gantzer et al.*, 2009]. *Gantzer et al.* [2009] developed the correlation after discovering that SOU increased proportional to gas flow rate in SHR, which was due to increased turbulence and mixing caused by diffuser operation. However, the correlation may overestimate the local SOU at a longitudinal location away from the diffuser, where the sediments are less impacted by diffuser-induced mixing.

3.3. Effects of Bubble Plume on Lake Circulation and Dissolved Oxygen Field

[31] Bubble plume circulation and mixing were examined using simulation results in which the only forcing was due to diffuser operation (i.e., no inflows or outflows, SOU, or atmospheric forcing). The initial temperature and gas flow rate and composition were equal to the 1998 diffuser test.

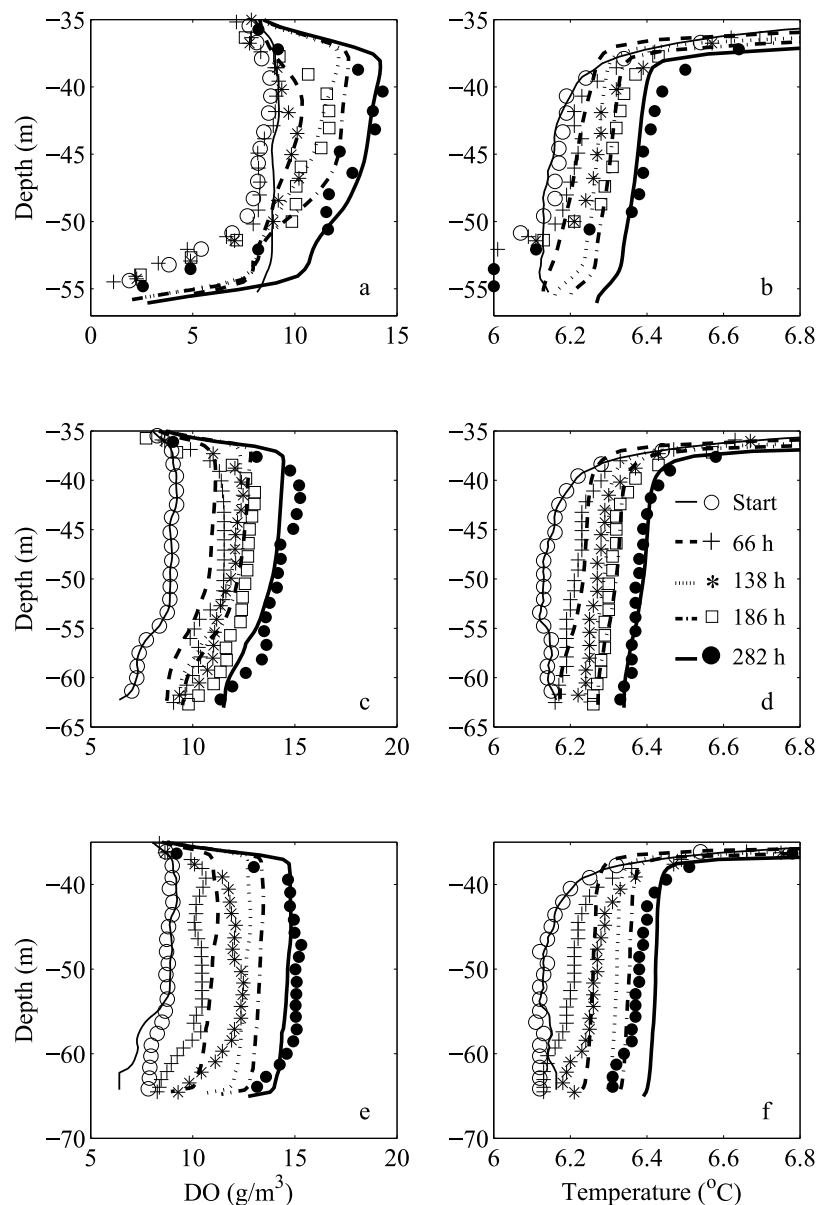


Figure 5. Hypolimnetic profiles predicted by coupled bubble plume–3-D hydrodynamic model, represented as solid and dashed lines. Observed DO and temperature are represented as symbols. Data were collected from Spring Hollow Reservoir, Virginia, United States, during linear diffuser operation in August 2003 and correspond to (a and b) 530 m upstream of dam (about 300 m upstream of diffuser end), (c and d) 230 m upstream of dam (midpoint of diffuser), and (e and f) 8 m upstream of dam (about 70 m downstream of diffuser start).

The lake was assumed to have zero initial oxygen concentration to track the fate and transport of oxygen added by the diffuser. The duration of the simulation was set to 1 day.

[32] Figure 6a shows that two nonsymmetric circulation cells developed laterally on either side of the diffuser: water flows away from the diffuser at the top of the plume (detrainment), but it flows toward the plume below the top (entrainment). Simulations conducted in a rectangular basin with the diffuser located in the center yielded circulation cells which are symmetric laterally on both sides of the diffuser. This suggests that the asymmetric circulation in SHR is due to the irregular bathymetry.

[33] The predicted vertical velocity of the water directly over the diffuser is negative (Figure 6a), which appears to be inconsistent with the sign of the basin-scale lateral recirculation cells that develop on both sides of the diffusers. In the coupled model, the motion of water within the plume is simulated as a relocation of water parcels occurring instantaneously. The model, in turn, captures the response of the near field (the water column above the diffuser and the water columns adjacent to this column) to that rapid relocation process. This response consists of replacing part of the plume-relocated water with water from cells above and with water entrained from the ambient. These results are consistent with the velocity observations collected in Lake

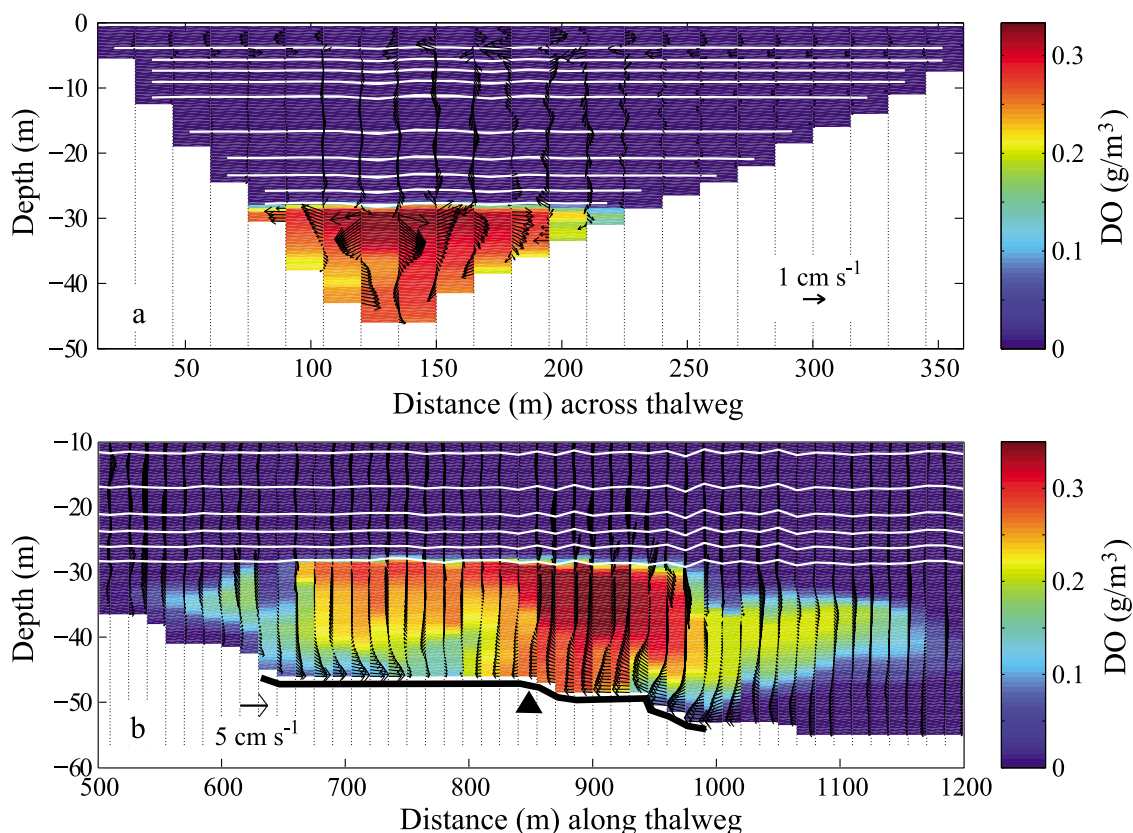


Figure 6. Predicted planar velocity (u , w), temperature, and oxygen fields at (a) a lateral cross section and (b) a longitudinal cross section in Spring Hollow Reservoir, Virginia, United States. The results correspond to 12 h after the start of linear diffuser operation in September 1998. The velocities are shown as arrows, the white lines are isotherms represented every 1°C, and the DO concentration is shown in color. The distance is measured from a point located at the southwest corner of the computational domain. The distance across the thalweg increases from west to east, while the distance along the thalweg increases from south to north. The solid black line along the bottom in Figure 6b marks the location of the diffuser, and the solid triangle defines the position along the thalweg for the cross section in Figure 6a.

Hallwil, Switzerland by McGinnis *et al.* [2004], which indeed reveal downward flowing water near the diffuser due to plume near-field recirculation. Water relocated from near the bottom will consist of ambient far-field water flowing (from neighboring water columns in the model) toward the diffuser as well as downward flowing water from the near field (i.e., from the same water column affected by the relocation process in the coupled model).

[34] The plume not only induces hypolimnetic currents, but it also causes the water above the hypolimnion to move (Figure 6). The horizontal velocity above the hypolimnion oscillates in the vertical direction. The oscillatory nature (in space) of the velocity field is likely related to the occurrence of diffuser-induced internal oscillations in the lake thermal structure. These oscillations are revealed by the isotherms being displaced from their horizontal equilibrium position (Figures 6a and 6b), and appear to be of small amplitude and large wave number. The development of such internal oscillations characterizes the transitory response of a stratified water body to external perturbations [Fischer *et al.*, 1979]. They occur, for example, after the initiation of withdrawal operations in stratified bodies [Imberger *et al.*, 1976; Imberger, 1980; Silvester, 1980]. These internal oscillations

are weak and, due to their high wave number, should rapidly dissipate [Münich, 1996]. Hence, the oscillations might not be observable in the field, where they could be masked by currents or wind-induced internal waves, which are larger in magnitude and appear to dominate the observations in field data sets collected in other reservoirs [Vidal *et al.*, 2005, 2007].

[35] The distribution of DO added by a hypolimnetic oxygenation system is of great interest to reservoir managers and design engineers. The coupled model predicts a region of highest DO directly above the linear diffuser (Figure 6b). Away from the diffuser and along the lake thalweg, the relative change in DO is less. (The presence of DO laterally (Figure 6a) and longitudinally (Figure 6b) away from the diffuser is due to spreading of added DO by plume-induced mixing in the hypolimnion.) Downstream of the diffuser and toward the dam, the hypolimnion appears to be more oxygenated than upstream. The topography and slope of the reservoir bottom may have contributed to this effect. The rise height of the plume (i.e., vertical distance from the diffuser to the top of the hypolimnion) is greater at the downstream end of the diffuser compared to the upstream end because of the deeper diffuser depth (Figure 6b). The greater plume rise

height results in more entrainment and detrainment and, subsequently, enhanced circulation and mixing, which contributes to greater distribution of DO downstream of the diffuser compared to upstream.

[36] Upstream and downstream of the ends of the linear diffuser, the maximum DO concentrations appear not at the top of the hypolimnion but at middepth (Figure 6b). The predicted longitudinal circulation away from the diffuser appears to be focused not at the top of the hypolimnion, but below it. This motion could be mainly driven by operation of the diffuser, but the baroclinic pressure gradients created by the diffuser in the hypolimnion also play a role. The diffuser rapidly mixes the hypolimnetic water lying above it and generates temperature gradients along the reservoir thalweg (Figure 6b). The velocity scale U of density driven currents can be estimated as [Rueda et al., 2003a]

$$U \approx \sqrt{\beta \Delta \theta g H} \quad (10)$$

where β is the thermal expansion coefficient ($\approx 1.4 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$), $\Delta \theta$ is the magnitude of the temperature differences, and H is a length scale representing the depth of the water column. For $\Delta \theta \approx \text{O}(10^{-1} \text{ }^\circ\text{C})$ and $H \approx \text{O}(10) \text{ m}$ as observed in SHR (Figure 6b), U is $\text{O}(10^{-2}) \text{ m s}^{-1}$, as predicted by the model (Figure 6b).

3.4. Sensitivity Analysis

[37] The parameters selected for the sensitivity analysis are either difficult to measure (SOU and entrainment and detrainment velocity), can be controlled through system design or operation (diffuser length L_{diff} , gas flow rate Q_g , and initial bubble radius r_{ini}) or relate to the accuracy with which the reservoir bathymetry is known. Sediment oxygen uptake, for example, is generally modeled using a reservoir-wide constant coefficient (k_{sou}), the value of which is subject to a large degree of uncertainty. The SOU range for lakes is typically $0.06\text{--}2 \text{ g m}^{-2} \text{ d}^{-1}$, as reported by Chapra [1997]. Similarly, Cole and Wells [2006] stated that SOU generally ranges from 0.1 to $1.0 \text{ g m}^{-2} \text{ d}^{-1}$, but can be higher. Here, the SOU was varied between $\pm 10\%$ of the baseline value used in the simulations of the 1998 experiment, in accordance with the conditions required by the FOVA.

[38] For the sensitivity analysis, the velocity of the entrainment or detrainment flow was assumed initially to be equal to the velocity of the ambient flow, or U_{amb} . Other values are also plausible (refer to section 2.4), but we do not have any a priori arguments that justify one value. One simulation was conducted with the entrainment and detrainment velocities set equal to the velocities through the cell flux faces (U_f), to determine the effects of momentum on predicted temperature and DO fields.

[39] Other parameters included in the sensitivity analysis (diffuser dimensions, gas flow rate, and, to a lesser extent, initial bubble size) can be controlled through diffuser design and operation. The sensitivity range for L_{diff} corresponds to about $\pm 8\%$ of the baseline value used in the simulations of the 1998 experiment, which is equal to the length of two columns in the grid. Similar to SOU, the ranges for L_{diff} and initial bubble radius are $\pm 10\%$ of the baseline value.

[40] Reservoir bathymetry can also be associated with a large degree of uncertainty. During the process of converting topographic data to a bathymetry input file for a

hydrodynamic model, the effective volume-depth curve, and consequently the transport and mixing processes that occur within that domain, can be altered. Uncertainty in model bathymetry not only results in changes to the volume of water receiving oxygen, but might also affect the depth of the diffuser, modifying the diffuser operating conditions. In this work, we focus on bathymetry effects that change the lake volume but not the diffuser depth. The bathymetry in the sensitivity simulations was modified either by increasing the depth of the deeper columns (those with depths greater than 35 m) in the hypolimnion by 1.5 m and making the shallower columns even shallower, or, by decreasing the depth of the deeper columns by 1.5 m and deepening the shallower columns. The changes introduced in the depths of the shallower columns of the hypolimnion were made so that hypolimnetic volume was conserved. The depth-volume curves were, in turn, modified. Modifying the depths of the water columns in the hypolimnion resulted in about 10% of the total volume of the hypolimnion being shifted to different depths for each perturbation condition.

[41] Of all the parameters analyzed, the largest source of uncertainty for both the DO and temperature predictions is the model bathymetry, which accounts for about 90% of the total uncertainty (Table 4). This is consistent with the results of Blumberg and Georgas [2008], who found that the sensitivity coefficients for bathymetry were usually much higher than the coefficients for wind forcing or tidal amplitude in a three-dimensional estuarine model. Modifying the grid resolution would likely affect the magnitude of the uncertainty associated with the bathymetry, but the bathymetry would probably still be the greatest contributor to the total model uncertainty. These results suggest that special care should be taken to acquire accurate bathymetric data, and to generate model grids that represent the water body's area distribution with depth as closely as possible.

[42] The parameter with the lowest sensitivity coefficient for DO is the initial bubble size (Table 4). Oxygen transfer efficiency is a strong function of initial bubble size for radii greater than about 10^{-3} m [Singleton et al., 2007]. However, we assumed 100% oxygen transfer efficiency for all of the simulations in this work, which is why the coupled model is insensitive to initial bubble size. Of the remaining three parameters, the gas flow rate has the second largest sensitivity coefficient for both predicted temperature and DO (Table 4). By increasing the gas flow rate, water managers increase the rate of oxygen addition in the hypolimnion, but they will also increase the rate at which warmer and potentially lower-oxygen water above the thermocline (due to metalimnetic DO minima [Effler et al., 1998; Schram and Marzolf, 1994; Shapiro, 1960]) is entrained into the hypolimnion (as evidenced in Table 4 by the relatively large sensitivity of temperature to gas flow rate). Higher hypolimnetic temperatures and increased water velocity above the sediments may lead to increased SOU, which would partially negate the oxygenation effects of increasing the diffuser gas flow rate.

[43] Our analysis indicates that determining accurate values of SOU is not critical to obtaining good DO model predictions. However, given that rates of areal hypolimnetic oxygen demand (which are commonly assumed to approximate SOU rates) reported in the literature can vary by up to 2 orders of magnitude [Chapra, 1997], the relative scale of the value should be well defined. The insensitivity of pre-

Table 4. Sensitivity Coefficients S Estimated for Coupled Hydrodynamic–Bubble Plume Model Application to Spring Hollow Reservoir, Virginia, United States^a

Model Input Parameter	Baseline Value, p_0	Perturbation, $p_0 \pm \Delta p$	RMSE _{temp}	RMSE _{DO}	$\frac{\Delta p}{p_0} \times 100$ (%)	$\Pi(\text{temp})_{11\text{d}}$	$\Pi(\text{DO})_{11\text{d}}$	S_{temp}	S_{DO}
SOU ($\text{g m}^{-2} \text{d}^{-1}$)	0.6	0.54	0.00016	0.056	−10	0.0000091	0.0066	0.000091	0.066
	0.6	0.66	0.00033	0.054	10	0.000016	0.0065	0.00016	0.065
Entrainment/detrainment momentum	U_{amb}	$U_f/2$	0.34	0.23	NA	0.020	0.029	NA	NA
	U_{amb}	U_f	0.78	0.47	NA	0.043	0.057	NA	NA
Diffuser length (m)	360	330	0.073	0.055	−8	0.0039	0.0035	0.047	0.042
	360	390	0.026	0.028	8	0.0012	0.0022	0.014	0.026
Gas flow rate ($\text{Nm}^3 \text{h}^{-1}$)	43	39	0.050	0.14	−10	0.0024	0.017	0.024	0.17
	43	47	0.095	0.16	10	0.0048	0.019	0.048	0.19
Initial bubble radius (m)	5.7×10^{-4}	5.13×10^{-4}	0.010	0.011	−10	0.00044	0.00085	0.0044	0.0085
	5.7×10^{-4}	6.27×10^{-4}	0.020	0.021	10	0.00089	0.0021	0.0089	0.021
Deeper hypolimnion	NA	NA	0.11	0.099	1.62	0.0027	0.010	0.17	0.62
Shallower hypolimnion	NA	NA	0.19	0.11	1.62	0.0057	0.023	0.35	1.39

^aSensitivity of temperature (temp) and dissolved oxygen (DO) coupled model predictions determined for each parameter analyzed. RMSE is root-mean-square error; NA means not applicable.

dicted DO concentrations to changes in SOU is likely because the rate of oxygen addition from the bubble plume was four times the SOU rate. Note also that by changing the SOU, the temperature structure of the reservoir is altered to a small degree (Table 4). This suggests that oxygen levels could influence mixing dynamics in the lake. This apparently unrelated consequence can be explained by considering the rate at which rising bubbles dissolve, and hence, the rate at which buoyancy changes vertically along the plume rise height. Dissolution of the bubbles, which contain gaseous oxygen, is a function of the DO concentration difference between the interior of the bubble and the ambient water. Changes in the SOU can affect the ambient DO concentration, thereby affecting bubble dissolution, the plume rise height, and, subsequently, hypolimnetic mixing and erosion of the thermocline.

3.5. Momentum of Plume Entrainment and Detrainment

[44] There are three primary sources of uncertainty in hydrodynamic and transport modeling studies: (1) the structure of the model, (2) the use of parameters with unknown value (knowledge uncertainty), and (3) the intrinsic stochastic nature of hydrological and/or meteorological phenomena forcing the transport in aquatic systems (stochastic uncertainty) [Hession *et al.*, 1996; Melching, 1995; van Straten and Keesman, 1991]. A large source of uncertainty in the structure of the coupled model is the momentum or velocity assigned to plume entrainment and detrainment. At the center of the plume, the horizontal velocity components u_0 and v_0 of the water that has been entrained or detrained are equal to the ambient velocity U_{amb} . At the perimeter of the water column in which the plume is located, the horizontal velocity of the water being entrained or detrained is equal to the entrainment or detrainment flow rate divided by the area of the flux faces (U_f) (Figure 1). These two velocity values, and any other within their range, are plausible values for u_0 and v_0 . We will refer to the option of setting $(u_0, v_0) = U_{\text{amb}}$ as mode or configuration 1, or C1; and $(u_0, v_0) = U_f$ as mode or configuration 2, or C2. Note that in the model validation simulations discussed previously, (u_0, v_0) was set to $U_f/2$.

[45] Simulations of the 1998 experiments conducted with configurations C1 and C2 yielded different temperature and DO profiles near the dam at the end of the 11 day diffuser test (Figure 7). The root-mean-square difference (RMSD) of the temperature and DO profiles at the end of the 11 day period simulated with modes C1 and C2 was 0.78°C and 0.47 g m^{-3} , respectively (Table 4). These RMSDs are similar in magnitude to and greater than the RMSDs encountered when varying the bottom bathymetry, to which the model exhibits the largest sensitivity (as discussed previously). The choice of appropriate values of u_0 and v_0 is clearly crucial.

[46] The bulk DO concentration in the hypolimnion is less for C2 compared to C1 as a consequence of larger vertical entrainment rates of the more oxygen depleted water of the metalimnion into the hypolimnion (Figure 7a). Predicted hypolimnion temperature was also larger for C2 (Figure 7b) as a result of larger vertical entrainment rates of warmer metalimnetic water in the hypolimnion. The elevation of the top of the hypolimnion at the end of the simulations was higher for C2 compared with C1, which is indicative of a larger portion of the lake volume being part of the hypolimnetic volume. The larger hypolimnetic volume for C2 could have contributed to the lower predicted bulk DO in this configuration, compared with C1, by diluting the oxygen mass addition of the plume.

4. Conclusions

[47] To model the complex interaction between a bubble plume used for hypolimnetic oxygenation and the ambient water body, a bubble plume model for a linear diffuser was coupled with a three-dimensional reservoir hydrodynamic model. Overall, the coupled model predicted the bulk DO and temperature profiles well for the two diffuser tests, September 1998 (compressed air at $45 \text{ Nm}^3 \text{h}^{-1}$) and August 2003 (pure oxygen at $23 \text{ Nm}^3 \text{h}^{-1}$). The comparisons between predictions and observations generally improved as each diffuser test proceeded. The simulated model profiles became more uniform vertically as the run progressed compared to observations, which could have been caused by overestimated vertical mixing and/or the inappropriate selection of detrainment momentum for the coupled model. The detrainment momentum, which was assumed to be

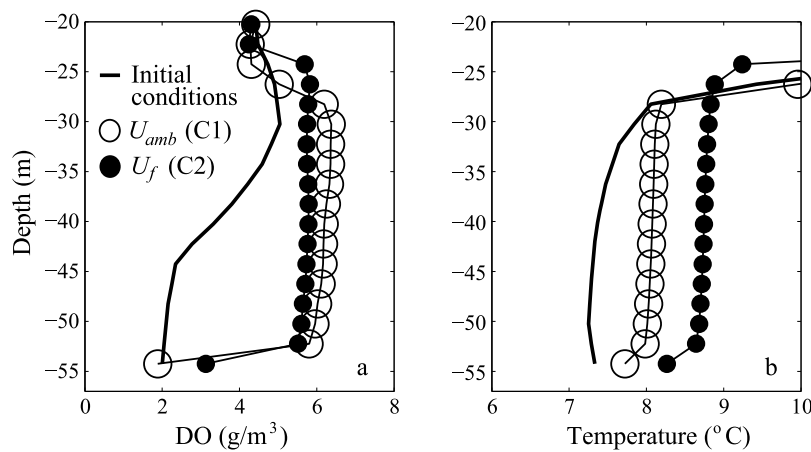


Figure 7. Comparison of effects of entrainment and detrainment velocity selection on hypolimnetic profiles predicted by coupled bubble plume–3-D hydrodynamic model after 11 days of linear diffuser operation in Spring Hollow Reservoir, Virginia, United States. Predictions are for the deepest point near the dam. U_{amb} (configuration C1) corresponds to velocities equal to ambient horizontal values, and U_f (configuration C2) corresponds to entrainment/detrainment velocities equal to Q/A_f .

proportional to $U_f/2$ for the baseline runs, may have been overestimated.

[48] Simulations to examine the effects of bubble plume operation on mixing and reservoir circulation revealed the development of two nonsymmetric circulation cells both laterally across the diffuser and longitudinally upstream and downstream of the diffuser: water flows away from the diffuser at the top of the plume (detrainment) and flows toward the plume below the top (entrainment). On the basis of the results of similar simulations conducted in a rectangular basin, it was concluded that the asymmetry of the predicted circulation in SHR is due to an irregular reservoir bathymetry. The plume rapidly mixes the hypolimnion directly above it and causes the water above the linear diffuser to be colder compared to away from the ends of the diffuser, which results in intrusions flowing from above the diffuser into the upstream and downstream water columns.

[49] Of all the parameters included in the sensitivity analysis, the largest source of uncertainty for both predicted DO and temperature is the model bathymetry, which accounted for about 90% of the total uncertainty. Therefore, special care should be taken when determining reservoir bathymetry as well as constructing the model grid; the hypsographic depth-area curves used in the model should match the actual bathymetry as closely as possible. One of the greatest sources of uncertainty in the structure of the coupled model is the momentum or velocity assigned to plume entrainment and detrainment. The horizontal momentum of the entrainment or detrainment can vary from the ambient value to the value at the perimeter through the flux faces. The horizontal plume-induced momentum significantly affects the predicted intensity of mixing and lake circulation. Consequently, the time scales of transport (for example, the time to reach uniform conditions in the hypolimnion) are also affected. Additional research on the plume detrainment momentum appropriate for the coupled model needs to be conducted.

[50] Coupling of the plume model with the hydrodynamic model is promising as an engineering tool for design and evaluation of hypolimnetic oxygenation systems. As a

complement to this, the coupled models can be used to more thoroughly investigate the effect of plume-induced mixing in a lake or reservoir.

Notation

- A dispersion coefficient for momentum, $\text{m}^2 \text{s}^{-1}$.
- C dissolved concentration, mol m^{-3} .
- c_p specific heat of water, $\text{J kg}^{-1} \text{ per } ^\circ\text{C}$.
- D dispersion coefficient for temperature and constituents, $\text{m}^2 \text{s}^{-1}$.
- DC detrainment cell, dimensionless.
- E entrainment, $\text{m}^2 \text{s}^{-1}$, or energy, J .
- EC entrainment cell, dimensionless.
- F model output variable for sensitivity analysis, units vary.
- F_D dissolved species flux, mol s^{-1} .
- F_G gaseous species flux, mol s^{-1} .
- F_S salinity flux, kg s^{-1} .
- F_T temperature flux, $^\circ\text{C m}^{-3} \text{s}^{-1}$.
- Fr Froude number, dimensionless.
- f body force, Coriolis parameter, s^{-1} .
- g gravitational acceleration, m s^{-2} .
- H Henry's constant, $\text{mol m}^{-3} \text{bar}^{-1}$; source of heat, $\text{J m}^{-3} \text{s}^{-1}$; or water depth, m .
- K_L mass transfer coefficient, m s^{-1} .
- L length, m .
- M water momentum, $\text{m}^4 \text{s}^{-2}$.
- N number flux of bubbles, s^{-1} .
- O dissolved oxygen concentration, g m^{-3} .
- P pressure, Pa .
- p sensitivity parameter, units vary.
- Q volumetric flow rate, $\text{m}^3 \text{s}^{-1}$.
- R relative contribution to total uncertainty, dimensionless.
- r bubble radius, m .
- S salinity, g kg^{-1} ; fluid source strength, $\text{kg m}^{-3} \text{s}^{-1}$; or sensitivity coefficient, dimensionless.
- T temperature, $^\circ\text{C}$, or characteristic time scale, s .
- t time, s .

U water velocity or velocity scale of density driven currents, m s^{-1} .
 u velocity in x direction, m s^{-1} .
 V volume of cell, m^3 .
 v velocity in y direction or velocity, m s^{-1} .
 W width, m , or work, J .
 w velocity in z direction, m s^{-1} .
 x length of grid cell in east–west direction, m .
 y plume gaseous concentration, mol m^{-3} , or length of grid cell in north–south direction, m .
 z depth, elevation, or length of grid cell in vertical direction, m .

Greek letters

α entrainment coefficient, dimensionless.
 β thermal expansion coefficient, $^{\circ}\text{C}^{-1}$.
 ζ free surface elevation, m .
 η efficiency
 θ temperature, $^{\circ}\text{C}$.
 λ spreading coefficient, dimensionless.
 ρ density, kg m^{-3} .

Subscripts

0 water added or removed at a source or sink cell, or sensitivity baseline.
 a , amb ambient.
 atm atmospheric.
 b bubble.
 bas reservoir basin.
 DO dissolved oxygen.
 d detrainment.
 $diff$ diffuser.
 e entrainment.
 f flux face.
 fin final.
 g gas.
 h horizontal.
 i gas species (oxygen or nitrogen).
 igw internal gravity wave.
 ini initial.
 iso isothermal compression.
 kin kinetic.
 $lake$ characteristic lake or reservoir.
 $mech$ mechanical.
 p plume water and gas mixture, or sensitivity parameter.
 plu plume.
 pot potential.
 SOU sediment oxygen uptake.
 v vertical.
 w plume water.

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J. C. Little and V. L. Singleton, Department of Civil and Environmental Engineering, Virginia Polytechnic Institute and State University, 418 Durham Hall, Blacksburg, VA 24061-0246, USA. (jcl@vt.edu)

F. J. Rueda, Instituto del Agua, Universidad de Granada, C/Ramón y Cajal, 4, E-18071 Granada, Spain.